

Marker-Less Micro Aerial Vehicle Detection and Localization Using Convolutional Neural Networks

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Abstract—A relative localization system for micro aerial vehicles (MAVs), which is able to work without any markers or other specialized equipment, is presented in this letter. The system utilizes images from an onboard camera to detect nearby MAVs using a convolutional neural network. When compared to traditional computer vision-based relative localization systems, this approach removes the need for specialized markers to be placed on the MAVs, saving weight and space, while also enabling localization of non-cooperating robots. The system is designed and implemented to run online, onboard an MAV platform in order to enable relative stabilization of several MAVs in a formation or swarm-like behavior, when operating in a closed feedback loop with the control system of the MAVs. We demonstrate the viability and robustness of the proposed method in real-world experiments. The method was also designed for the purpose of autonomous aerial interception and is a fitting complement to other MAV detection and relative localization methods for this purpose, as is shown in the experiments.

Index Terms—Aerial systems: perception and autonomy, field robots, multi-robot systems, recognition.

I. INTRODUCTION

A NOVEL marker-less relative localization system for nearby micro aerial vehicles (MAVs) is presented in this letter. Relative localization of robots is a necessary prerequisite for solving many problems involving robot cooperation, such as swarm-like behavior, compact formations of autonomous robots, cooperative exploration, etc. Relative localization is less restrictive than conventional absolute localization methods, if measuring the absolute position of the robots is not required. Absolute localization methods usually rely on existing infrastructure, e.g. a global system of satellites (GNSS), or on expensive pre-installed camera set-ups in the case of motion capture systems. Relative localization enables more flexible usage in areas where absolute localization systems are not available.

The relative localization method presented in this letter further relaxes the requirements on the robots that are being localized,

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Fig. 1. Autonomous aerial interception performed by our AAIS platform. The intruder MAV (left) is localized using a marker-less relative localization system and autonomously captured by the AAIS MAV (right) using an onboard net launcher. From top to bottom: Interception before launching the net ($t = 0$ s), interception after launching the net ($t = 0.5$ s), the intruder is caught in the net ($t = 1$ s). Video from the interception is available online.¹

it removes the need for the targets to carry specialized markers (as in [1]–[5]). This is important especially for MAVs, which have a limited carrying weight and volume capacity. Relative robot localization without the need for artificial markers also enables localization of non-cooperating robots, and is a vital prerequisite for realizing autonomous aerial intercepting systems (AAIS). As was discussed in our previous paper on this topic [6], autonomous aerial interception of non-cooperating aerial vehicles, e.g. commercially available hobby drones intruding a restricted area, is an important research topic (see Fig. 1).

The proposed relative localization method makes use of a convolutional neural network (CNN) for detecting the targets in a camera image. The detection is performed online on the MAV onboard computer and with reliability that allows the data that is obtained to be used in a control feedback for MAV group stabilization, and also for an AAIS system. The CNN is trained to detect MAVs and reports their bounding rectangles in the image (see Fig. 2). Given a known physical size of the MAVs (in the case of cooperating aerial swarms and formations), the distance of each detected MAV may be inferred from the size of the respective bounding rectangle. If a calibrated projection model of the camera is available, the direction of a detected MAV may be calculated by projecting the center of the bounding rectangle

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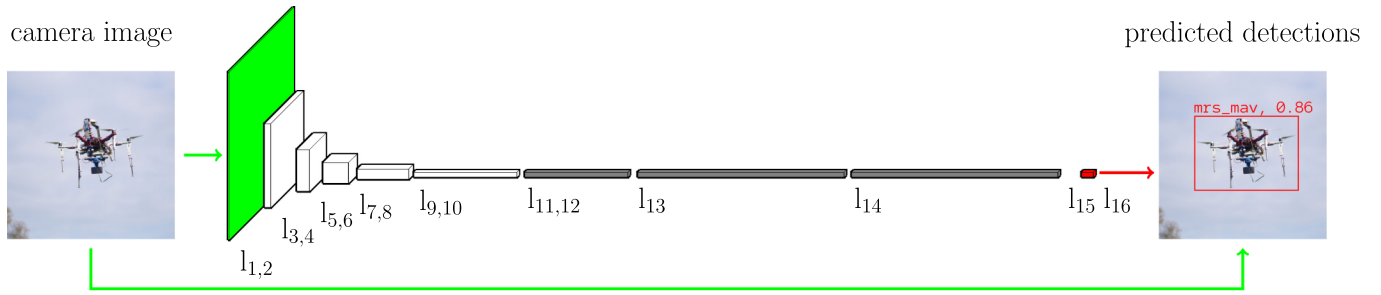


Fig. 2. A schematic of the convolutional neural network architecture used for MAV detection in this work. Input of the CNN is an RGB image. Layers l_1 to l_{12} are alternating convolutional and max-pooling layers, followed by three convolutional layers (l_{13} to l_{15}) and a final detection layer (l_{16}). Output of the CNN is a set of object candidates in the image with the corresponding bounding rectangles, class predictions and candidate confidences.

from the 2D image to a 3D space. By combining the distance and direction information, it is possible to obtain the position of the detected MAVs in the 3D space, relative to the camera origin. This is crucial for multi robot interaction.

A. Related Work

There are existing specialized systems for detecting amateur off-the-shelf MAVs using various methods, most of which are targeted at preventing breaches of no-fly zones [7]–[9]. However, the detection and localization methods used in these systems, such as radars, acoustic sensors or thermal cameras, are not suitable for relative localization onboard MAVs because of their weight, size or price. In comparison with the mainly static and ground-based sensors discussed in [7]–[9], a typical camera required for visual relative localization is a much lighter, smaller and cheaper sensor. This makes it suitable for use onboard an MAV.

Most visual relative localization systems for robots use artificial visual markers on the objects to be localized. These markers are detected in a camera image, and their shape, size and position in the image is used to calculate their 3D position relative to the camera. Numerous systems based on this general method have been used in various robotic applications [1], [2]. Representative examples of such systems are [3], [4], which use black and white markers, and offer scalable 3D relative localization with up to centimeter precision. Such systems are commonly used in the robotic community for various tasks requiring relative localization or identification. A slightly different approach, which makes use of ultra-violet (UV) light emitters as active markers, is presented in [5]. A combination of optical filters and a UV-sensitive camera sensor is used to detect the UV markers and filter out other light sources, as was shown in several experiments, demonstrating the practical usability of the system.

In this letter, we are focused on visual marker-less MAV relative localization, as it is much more broadly applicable with no requirements on hardware modification of the robots to be localized. Traditional computer vision approaches, such as template matching or detecting motion discrepancies caused by MAVs in a camera image, are used for marker-less visual MAV detection in [10] and [11]. However, in general object detection tasks, the traditional approaches have mostly been surpassed in accuracy and robustness by new methods based on

convolutional neural networks (CNNs), such as the YOLO [12], Faster-RCNN [13] and SSD [14] object detector CNNs. In [15], the authors describe their winning solution to the 2017 Drone-vs-Bird detection challenge, which utilizes a region-proposal algorithm to detect flying objects and a CNN to discriminate between MAVs, birds and other objects. Similarly, a CNN trained on an artificial dataset is evaluated for drone detection in [16]. Another approach utilizing a CNN is presented in [17], which introduces a novel variation of a semantic segmentation network to be used on embedded computers onboard MAVs, and the authors demonstrate the performance of their network on two datasets. These papers deal with visual detection, but none of them deal with estimating the relative position of the detected object (relative localization), or with integrating the results into a control feedback.

Existing research on marker-less relative localization is mostly focused on autonomous driving [18], [19] and usually assumes computational resources, which are unavailable onboard MAVs. Works which tackle this problem in the domain of MAV detection are relatively sparse [20], [21]. A relative localization system is presented in [20], where the authors combine a cascade AdaBoost detector with visual tracking and a probability hypothesis density filter to localize MAVs in an indoor environment. The MAV distance is estimated from the width of the bounding rectangle of the detection, but the relation is only determined empirically and is not analytically derived. A different approach is presented in [21], where the distance estimate is provided from the disparity between detections of the same MAV in two stationary cameras arranged in a stereo configuration. In contrast with the state-of-the-art in marker-less relative MAV localization, we demonstrate the practical usability of our system under realistic conditions in outdoor real-world experiments, where the system is applied in a closed loop with the MAV control algorithms.

B. Contributions

The method presented in this letter is able to localize robots without relying on markers, using only information from a monocular camera image. This means that it can be used in scenarios where it may not be expected that the robot being localized will carry any kind of markers or will cooperate in any other way. It also removes the need for extra hardware to be

carried by the robots, which may be important for example in the case of MAVs, which have limited carrying capacity.

In contrast to most state-of-the-art marker-less MAV detectors, our method is able to provide not only detections, but also relative positions of the detected MAVs in the camera image, and the localization runs online, onboard a flying MAV. To the best of our knowledge, this letter presents the first CNN-based solution of onboard marker-less relative localization of MAVs that are sharing the same working space, in control feedback for MAV coordination. This ability is demonstrated in two scenarios conducted in real-world conditions, leader-follower control and aerial drone interception, which both require a fast, precise and reliable sensory response (see Section III).

When compared to our previous work on this topic [6], a comparable performance is achieved by combining the two approaches, while using a sensor that is much smaller, cheaper and weighs less than 10% of the sensor used in [6], making it significantly more suitable for onboard use with MAVs. This is demonstrated in the experimental evaluation presented in Section III-B.

Moreover, the dataset used for training the neural network in this work is made publicly available. To the best of our knowledge, this is the only publicly available labeled dataset for MAV detection at the time of writing of this letter. This is another valuable contribution for the scientific community that could accelerate research in the application domains tackled in this letter (multiple MAVs sharing a common workspace and unauthorised drone interception), which have both gained increasing importance in the fast growing drone market.

II. MARKER-LESS RELATIVE LOCALIZATION

The input for the presented method is an RGB image stream originating from a camera. The output is a set of positions of all MAVs detected in the images, relative to the camera.

A convolutional neural network (CNN), designed for object detection, sequentially processes the images to output detection candidates. The output of the CNN is a set of predicted bounding rectangles of objects in the image and the corresponding confidence score for each bounding rectangle.

The bounding rectangle of each detection is used to estimate the relative position of the corresponding MAV. The size of the bounding rectangle is used to estimate the distance of the MAV based on its known physical dimensions. The direction of the MAV is determined from the position of the bounding rectangle in the image. The direction and distance information are combined to obtain an estimate of the position of the detected MAV relative to the camera.

A. Convolutional Neural Network Architecture

In this work, we did not focus on proposing a new CNN architecture, but on designing a relative localization system and integrating it with the rest of our platform. For this reason we leveraged an existing CNN architecture, and based our detector on the tiny-YOLO object detection CNN [12], which was modified for detecting a single class of objects. This architecture was chosen because it provides a high detection rate and low

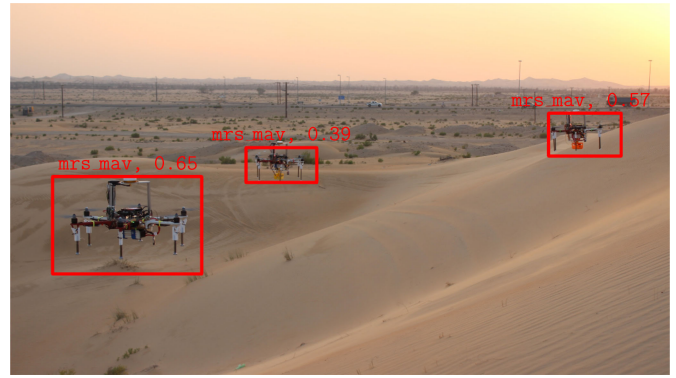


Fig. 3. Example output of the neural network in an image from the testing part of the dataset.

delay, while keeping sufficient robustness for our purposes, in comparison with more complex CNNs, such as YOLOv2 [12], Faster-RCNN [13] and SSD [14].

The tiny-YOLO is a state-of-the-art object detection CNN, which consists of standard convolutional and max-pooling layers to extract features from the input image, and a special fully-connected detection layer. The output of the detection layer contains candidates for detected objects, the confidence of the candidates, and their bounding rectangles in the image (see Fig. 3). Candidates with low confidence are discarded, and the rest is considered a valid output of the detection. A general scheme of the neural network is presented in Fig. 2.

The CNN was implemented using the Darknet neural network framework [22] and was integrated with the Robot Operating System [23]. The implementation utilizes an integrated GPU of the MAV onboard computer (namely the Intel Iris 6100) to achieve a higher processing rate, resulting in a processing time of 250 ms. This rate proved to be sufficient for basic practical applications, although it could be further improved e.g. by using a dedicated embedded hardware accelerator, such as the Nvidia Jetson TX2, if required by the application.

B. Training the Neural Network

The CNN was trained using the Stochastic Gradient Descent algorithm [24] with momentum. The weights of the layers l_1 to l_{13} were initialized with pre-trained weights from the Pascal VOC 2007 dataset [25] to speed up the training process. The training utilized a new dataset that was created specifically for the purpose of MAV detection. This dataset consists of manually labeled images of MAVs in different environments and lighting conditions, which were taken during our long-term research on multi-MAV systems working in real-world conditions [26], [27]. Images in the dataset were split into a training part and a testing part (see Fig. 4). During training, the images were augmented using standard methods, such as random cropping, rotation, gamma shifting, etc. A momentum of 0.9 and a learning rate of 10^{-4} were used. The CNN model was trained until the IoU (Intersection over Union) and the recall score of the model stopped converging on the testing data (after approximately $5 \cdot 10^4$ training batches with batch size 64). The Darknet



Fig. 4. The top two images are examples from the training dataset and the bottom two are from the testing dataset.

framework [22] was used for training. The training dataset is made publicly available and may be obtained online.²

C. Distance Estimation

The process for estimating the distance between a detected MAV and the camera is described in this section. It is based on the assumption that the bounding geometrical primitive of an MAV is a sphere with a known radius. For the case of autonomous interception where the size of the target may not be known apriori, the class estimation capabilities of the YOLO CNN may be leveraged to pick the most likely model from the set of commercially-available drone models with known dimensions. Another option is to use an RGB-D sensor (such as Intel Realsense D435), which provides aligned RGB and depth data, and the depth information is used to estimate the distance of the target. An apriori known target size is assumed further in this section.

The distance l from the origin of the camera coordinate system O_c to the center of the MAV C is estimated from angle α which is the angle between vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are directional vectors of 3D lines intersecting with the camera frame origin O_c and the centers of the vertical edges of the bounding rectangle of the detected MAV (see Fig. 5(a)). These lines are presumed to be tangent to the MAV bounding sphere S_{MAV} , as is depicted in Fig. 5(b). Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are obtained by projecting the corresponding undistorted pixel coordinates to 3D rays using a projection matrix of the camera. In our case, the projection was facilitated by a pinhole camera model, and a radial polynomial distortion model was used to undistort the pixel coordinates.

Angle α between vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ is obtained as

$$\alpha = \arccos \left(\frac{\mathbf{a}_1 \cdot \mathbf{a}_2}{\|\mathbf{a}_1\| \|\mathbf{a}_2\|} \right). \quad (1)$$

The formula for calculating the distance l using the angle α is derived from Fig. 5(b). The relation

$$l = r \csc \left(\frac{\alpha}{2} \right), \quad (2)$$

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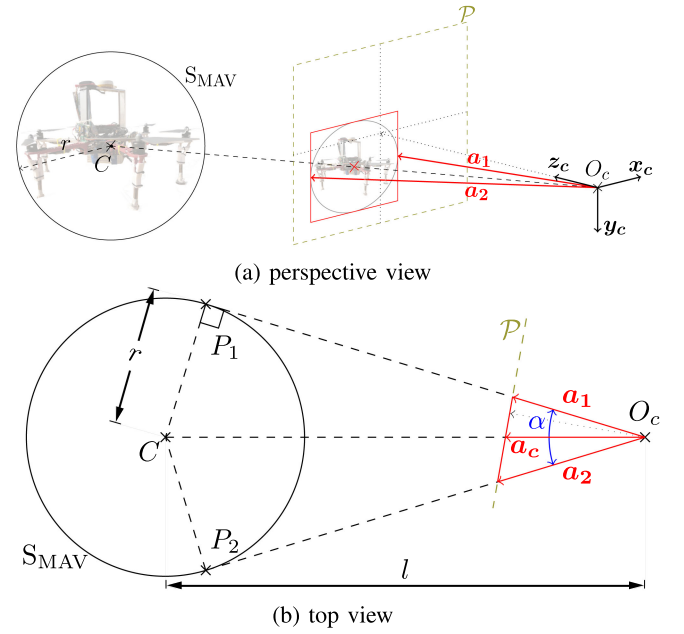


Fig. 5. Scheme of the projection of a detected MAV and its bounding sphere S_{MAV} to camera projection plane $\mathcal{P}$. A perspective view of the situation is shown in subfigure (a), and a top view (assuming that the camera is oriented horizontally) is shown in (b). The bounding sphere of the MAV (S_{MAV}) is specified by center point C and radius r . The camera coordinate frame is defined by the camera origin O_c and vectors $\mathbf{x}_c$, $\mathbf{y}_c$ and $\mathbf{z}_c$. Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are directional vectors of lines originating in O_c and intersecting the centers of the vertical edges of the bounding rectangle of the MAV projection to $\mathcal{P}$. These are assumed to be tangent to S_{MAV} . Points P_1 and P_2 are the respective intersections of these lines with S_{MAV} . The remaining variables are explained in Section II-C.

where $\csc(x)$ is the cosecant trigonometric function, can be derived from the right-angled triangle $\triangle O_c P_1 C$. The relation between vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ and distance l is obtained by combining equations (1) and (2) and by simplification using trigonometric identities in the form

$$l = \frac{\sqrt{2}r}{\sqrt{1 - \frac{\mathbf{a}_1 \cdot \mathbf{a}_2}{\|\mathbf{a}_1\| \|\mathbf{a}_2\|}}}. \quad (3)$$

Equation (3) describes the relation between the distance of the detected MAV from the camera origin and the width of the respective bounding rectangle.

D. Position Estimation

The position of a detected MAV in 3D is estimated using vectors $\mathbf{a}_1$, $\mathbf{a}_2$ and distance l , which are calculated as described in the previous section (see Fig. 5). A vector $\mathbf{a}_c$, pointing from the camera origin O_c towards the center of the detected MAV C , is obtained as the average of $\mathbf{a}_1$ and $\mathbf{a}_2$:

$$\mathbf{a}_c = \frac{\mathbf{a}_1 + \mathbf{a}_2}{2}. \quad (4)$$

Point C is obtained using these values as

$$C = l \frac{\mathbf{a}_c}{\|\mathbf{a}_c\|}, \quad (5)$$

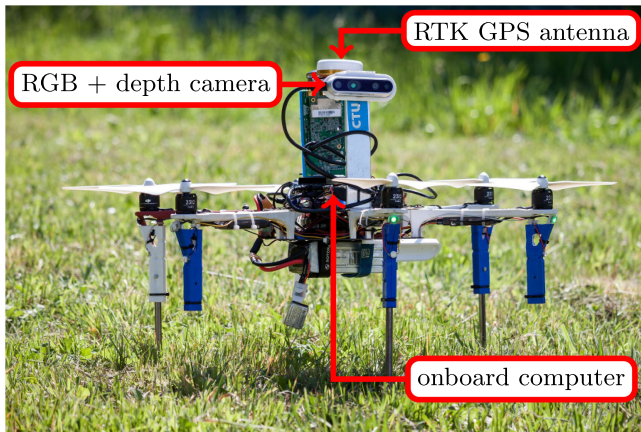


Fig. 6. The MAV platform used in the experiments. Note that for the relative localization method presented in this letter, the RTK-GPS and the depth data from the RGB-D camera are not necessary, and were only used for the experimental evaluation, as described in Section III.

which yields an estimate of the 3D position of the MAV in the camera coordinate frame.

E. Control Integration and Data Fusion

As demonstrated in Section III-A, outputs of the presented relative localization method may be used in a closed loop to control the position of MAV relative to the target. Specifically, the setpoint positions for the MAV that is being controlled are generated on the basis of the measured relative position of the target and the desired relative position, dictated by the application. The setpoint is then passed to an onboard Model Predictive Control tracker, which generates a trajectory for the MAV to achieve the setpoint, constrained by the MAV dynamics. The MAV is controlled to follow the generated trajectory by an SO3 controller. The trajectory tracking and the control systems are described in detail in [26], [27].

A multi-target tracking algorithm, presented in [6], was used to filter the detections. The tracking algorithm keeps a set of potential MAV tracks in the localization area. These are updated using the detections and a Kalman filter based on a model of MAV dynamics. This improves the detection precision by rejecting sporadic false positives, and it is a suitable framework for fusing measurements from different sources. It is leveraged in Section III-B.

III. EXPERIMENTS

Several real-world experiments have been conducted to evaluate the presented localization method under realistic conditions. A leader-follower experiment is presented in Section III-A as a representative example to demonstrate that the method may be used for MAV coordination in a dynamic scenario. Another experimental setup is presented in Section III-B, where the method is integrated into a system designed for autonomous aerial interception. Videos of the experiments are available at mrs.felk.cvut.cz/icra2020-mav-detection.

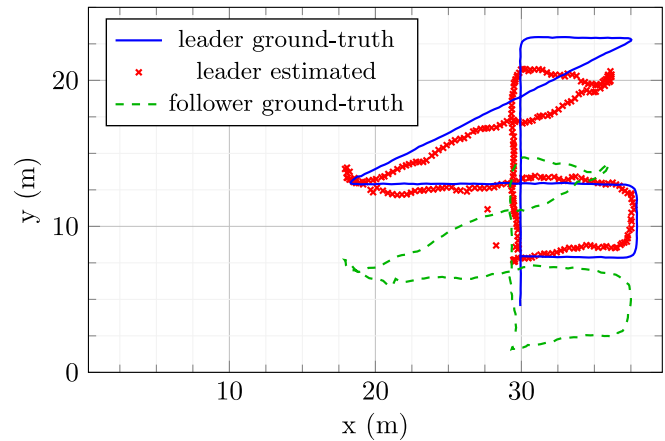


Fig. 7. Top view of the trajectories of the leader and follower MAVs, and the estimated trajectory of the leader during one of the experiments, described in Section III-A. Solid blue line represents ground-truth trajectory of the leader, red crosses show its estimated positions, and green dashed line represents ground-truth trajectory of the follower.

The experimental MAV platforms (see Fig. 6) used in the experiments carry an onboard camera and a computer for controlling the MAV and for running the neural network and the relative localization algorithm. Additionally, an RTK-GPS is used to provide precise ground-truth positions. For more details of the MAV platforms, see [26], [27].

A. Leader-Follower Experiment

In this experiment, a leader MAV flew along a predefined trajectory, and a follower MAV intended to keep a constant relative position with respect to the leader. The leader MAV used an RTK-GPS for self-localization and as a ground-truth reference for offline evaluation of the experiment. The follower used the RTK-GPS only to measure its ground-truth position, and it stayed in the formation with the leader based solely on the relative localization method presented here. The leader-follower experiment was conducted three times to ensure consistency of the results. The average localization error (the average Euclidean distance between the ground-truth positions of the leader MAV and the positions estimated by the relative localization system) during the experiments was 2.86 m.

The ground-truth positions of the leader and follower MAVs and the estimated positions of the leader during one of the experiments are plotted in Fig. 7. The corresponding localization errors over time are plotted in Fig. 8. Photos of the experiment are shown in Fig. 12. Based on a careful analysis of the data, it may be concluded that the main source of localization error is imprecise estimation of the distance of the target. This is caused by the unreliable size of the bounding rectangle, which is output from the CNN and is used to estimate the distance, as described in Section II-C. However, the precision, robustness and speed of the localization was adequate to realize an autonomous co-operative flight of two MAVs. The system can easily be used for multiple MAVs, as it allows multiple target detections (see Fig. 3).

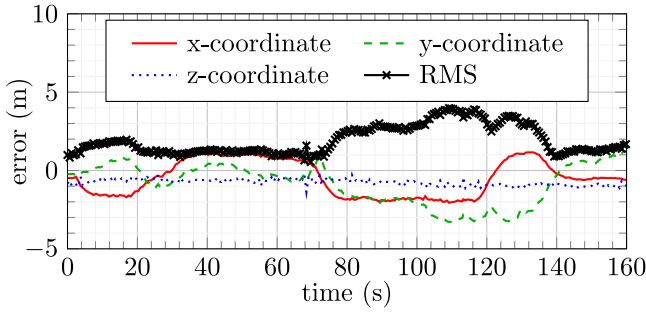


Fig. 8. Localization errors during the leader-follower experiment. The X, Y and Z errors are elements of the difference between the estimated position and the ground-truth position of the MAV. The values are in the static local coordinate frame, defined by the RTK-GPS. The RMS error is the Euclidean distance between the estimated and ground-truth positions.

TABLE I
COMPARISON BETWEEN DETECTION AND LOCALIZATION PERFORMANCE OVER THE WHOLE DATASET. VALUES IN PARENTHESES ARE WITHOUT FILTRATION

parameter	depth-based	CNN-based	combined
max. range	20 m	32 m	32 m
RMSE	2.41 m (2.65 m)	5.80 m (5.86 m)	3.76 m (4.46 m)
precision	99 % (97 %)	94 % (70 %)	96 % (79 %)
recall	77 % (77 %)	45 % (48 %)	90 % (89 %)

B. Combined Detection Experiment

A second experiment was performed to compare the proposed localization method with a different marker-less relative localization approach. One MAV carried a Realsense D435 sensor, which is an RGB-D (stereo) camera outputting aligned 1280 px $\times$ 720 px RGB and depth images at 30 Hz. A second MAV flew in an area visible by the RGB-D camera. The RGB images were used for the CNN-based localization presented in this letter, and the depth images were used for a depthmap-based localization method, which is described in [6]. Sporadic false positive detections were then filtered out using a tracking algorithm (also described in [6]). Since the system makes no assumptions on the number of MAVs being localized, a multi-target tracking approach was used. The parameters of the multi-target algorithm were the same as in [6], except for covariance of the distance estimate when using the CNN-based method, which was inflated according to its lower precision. Additionally, a combined approach was devised, fusing detections from both methods using the multi-target tracking framework.

Outputs of the different approaches were compared to obtain the data in Table I. The precision and the recall of the detection methods were obtained as

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (6)$$

where TP, FP and FN are the number of true positives, false positives and false negatives, respectively.

Based on the experiments, it may be concluded that position estimation using the CNN-based method is less accurate than the depthmap-based method. This is mainly caused by the imprecise estimate of the target distance, as discussed in Section III-A. The number of false positives of the CNN detector is higher, which

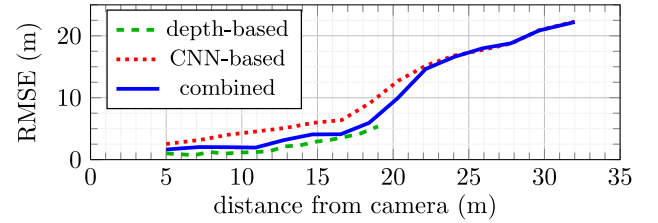


Fig. 9. Comparison of localization error over distance between the target and the camera for the different methods.

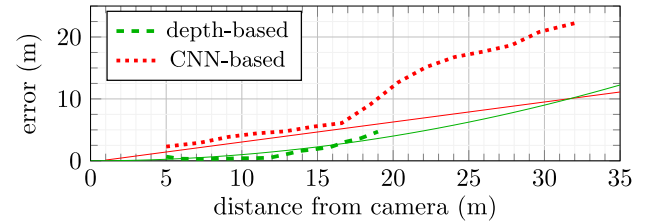


Fig. 10. Comparison of the expected and measured distance estimation errors. Expected errors are marked by solid lines of the corresponding color.

is reflected in lower detection precision (see Table I). However, this is mitigated by filtering using the multi-target tracking algorithm, and does not influence the performance of the control system significantly. The MAV is not robustly detected by the CNN when positioned against a structured background in the image (such as trees).

Nevertheless, in the remaining aspects, the CNN-based method has an advantage. Because of the small baseline of the D435 stereo camera (5 cm), the maximal usable range of the resulting depthmap is limited to approximately 20 m (a larger baseline cannot be applied mainly in the considered application of micro-scale MAVs). In contrast, the target MAV may be detected in the RGB image by the CNN at distances over 30 m. Additionally, the CNN-based method only requires a monocular camera, which further reduces the necessary sensory payload for the relative localization.

An analysis of the data shows that the relative speed of the target does not significantly influence the localization precision over the tested speed range (which ranged from 0 ms^{-1} to 3.5 ms^{-1}). The localization error grows with the distance of the target from the camera, as is evident from Fig. 9. After comparing Figs. 9 and 10, it may be concluded that the localization error is mostly caused by imprecise distance estimation,

An expected distance estimation error of the CNN-based method is plotted in Fig. 10. The expected error assumes an incorrectly predicted width of the bounding rectangles based on the IoU score of the CNN on validation data (which was 53%). The distance estimation error of the CNN-based method corresponds well to the expected error up to the range of 17 m, after which it grows rapidly (see Fig. 10). After analyzing the data, we conclude that the CNN has a bias towards a minimal size of the predicted bounding rectangles. Bounding rectangles of detected objects smaller than approximately 5% of the input image size (corresponding to a 17 m distance in case of the D435 camera and a 0.55 m MAV diameter, which were used in the experiment) are incorrectly predicted to be larger, biasing

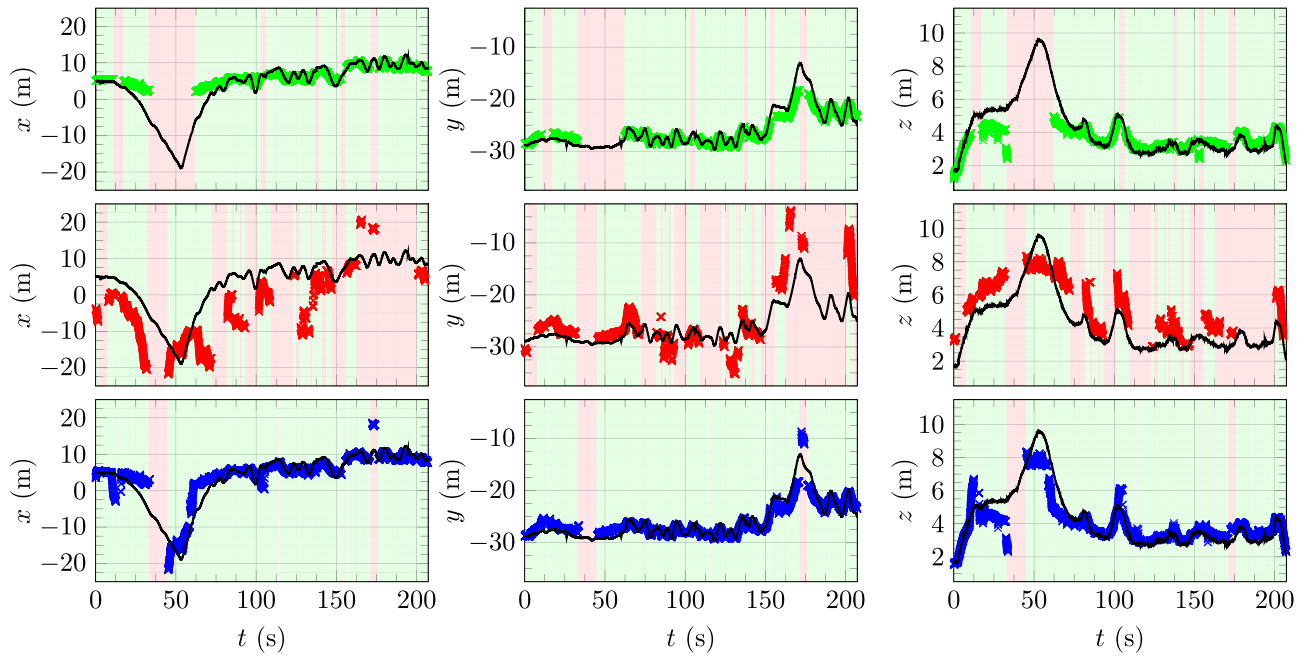


Fig. 11. Comparison of the localization results in separate coordinates using various detection methods during the experiment described in Section III-B. The values are in the static local coordinate frame, defined by the RTK-GPS, used for the ground-truth. The thick black line marks the ground-truth trajectory, and the colored crosses depict the localization results after filtering using the multi-target tracking algorithm from [6]. True positives are marked with a green background in the graph, and false negatives are marked with a red background. *Top row*: depthmap-based detection. *Center row*: CNN-based detection. *Bottom row*: combined detection.

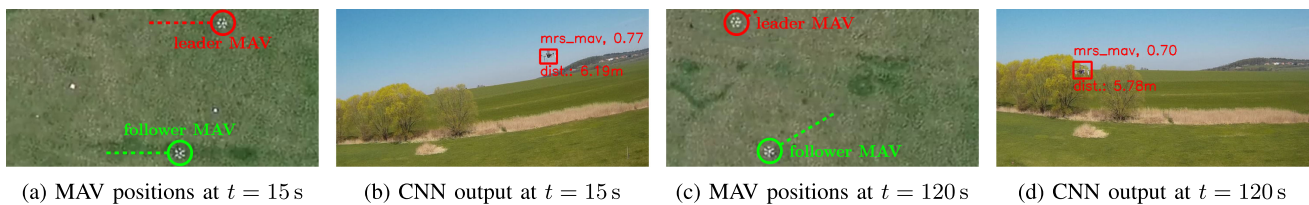


Fig. 12. Example photos and corresponding outputs of the CNN during the leader-follower experiment, described in Section III-A.



Fig. 13. Examples of detections and typical failure modes for the two compared detection methods. Detections from the CNN-based method are highlighted red and from the depthmap-based method green. (a): Both detection methods succeeded. (b): Failure mode of the CNN-based method (note that the MAV is hard to see even for a human because of the structured background behind it). (c): Failure mode of the depthmap-based method (the MAV is out of range of the stereo camera). (d): Both methods successfully detected the MAV, but the bounding rectangle is estimated poorly by the CNN, resulting in an imprecise position estimation of the target.

distance estimation of such objects. This bias may be caused by insufficient representation of small objects in the training dataset or it may be a limitation of the CNN architecture (such as insufficient input resolution). Distance estimation error of the depth-based method corresponds to the expected error based on mechanical and optical parameters of the D435 stereo camera and a 0.2 px subpixel precision of the stereo-matching algorithm.

The two methods complement each other well, and when combined they provide superior performance. If the parameters

are properly adjusted, the localization error is not significantly increased when detections from both methods are available (see Fig. 9 and Fig. 11). When one of the methods fails to detect the target, the other can keep updating the track and thus avoid false negatives, improving the recall by a significant amount when compared to using either of the methods alone (see Table I). A single stereo camera sensor can provide input for both methods, so the performance improvement does not require any additional hardware in comparison with using the depth-based approach

only. Fusing the two detection methods thus provides a suitable sensor for an AAIS platform, where both the precision and the recall of the detection are crucial parameters to ensure good performance of the system. Pictures of the experiments with examples of failure modes of both methods are shown in Fig. 13.

IV. CONCLUSION

This letter has presented a novel marker-less visual relative localization system suitable for MAVs. The system makes use of a state-of-the-art convolutional neural network (CNN) for detecting objects in a camera image, which was trained to detect MAVs. A directional vector from the camera origin to the detected MAV is obtained using camera projection, and the distance of the MAV is estimated from the size of the detected bounding rectangle, which is output from the CNN. The direction and distance together give the position of the detected MAV, relative to the camera, which can be used online in a control feedback.

The system is suitable for robotic tasks that require online relative localization of robots, such as swarming, formation flight or collision avoidance. It does not require any specialized sensors apart from an onboard camera, and it may be used in GPS-denied environments or in other situations in which conventional localization systems cannot be used, e.g. due to the limited payload of small MAVs or the presence of non-cooperating robots.

Real-world experiments have demonstrated that the presented system is robust. In addition, the proposed method was combined with our previous work on autonomous aerial interception, which is presented in [6]. The detection method presented in this letter complements our previous detection method well. Combining these two approaches significantly improved the performance of the drone interception system, as is discussed in Section III-B.

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